

Story5_Data608

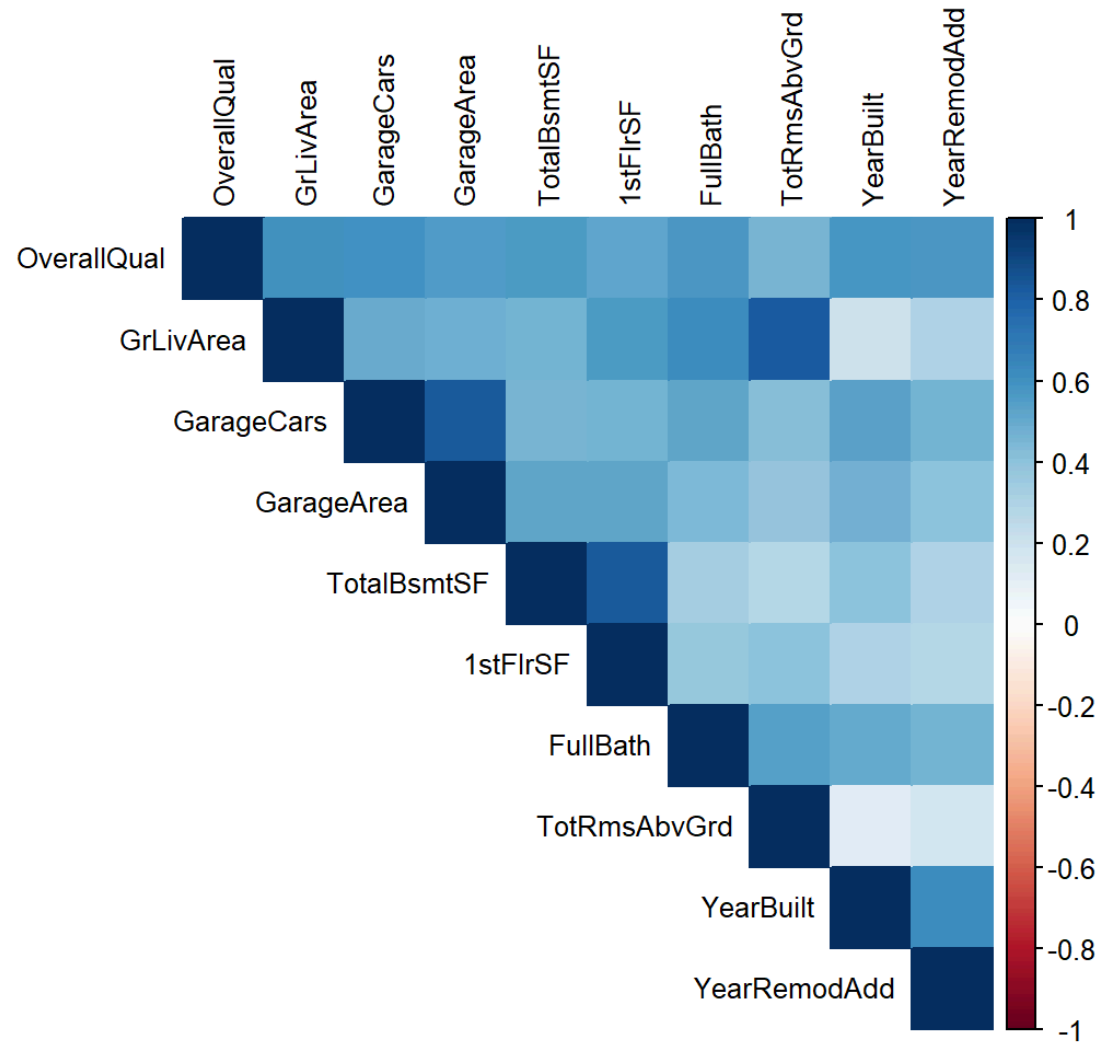
Mubashira Qari

2025-03-24

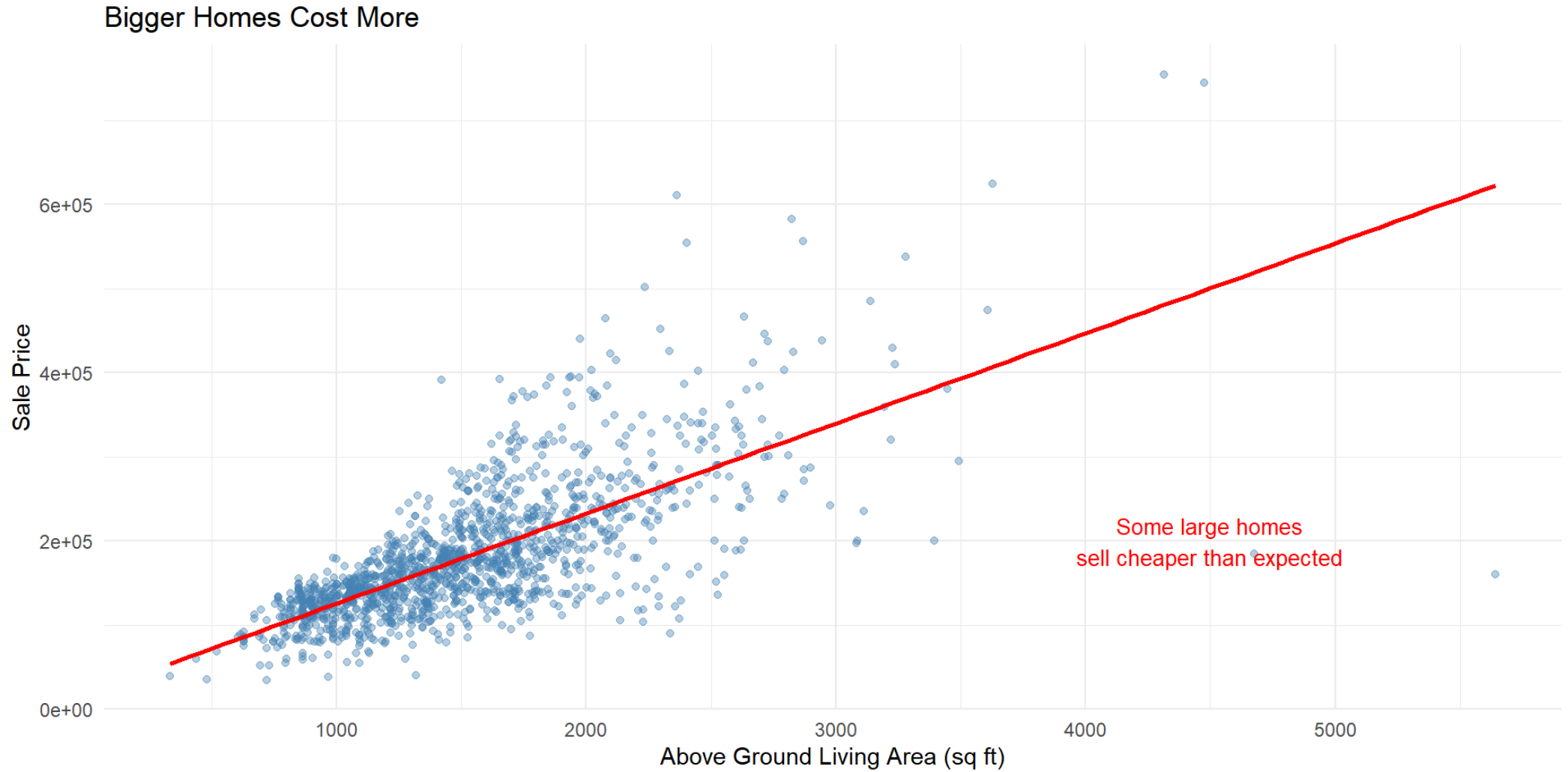
Defining Business Question

This week we will be building a model of the factors determining housing prices and using visualizations to explain their meaning and implications.

Sale Price Distribution & Top Correlations



Bigger Homes, Higher Prices



A Simple Yet Powerful Model Using Key Features

Call:

```
lm(formula = SalePrice ~ GrLivArea + OverallQual + GarageCars +  
    TotalBath + Age, data = train_df)
```

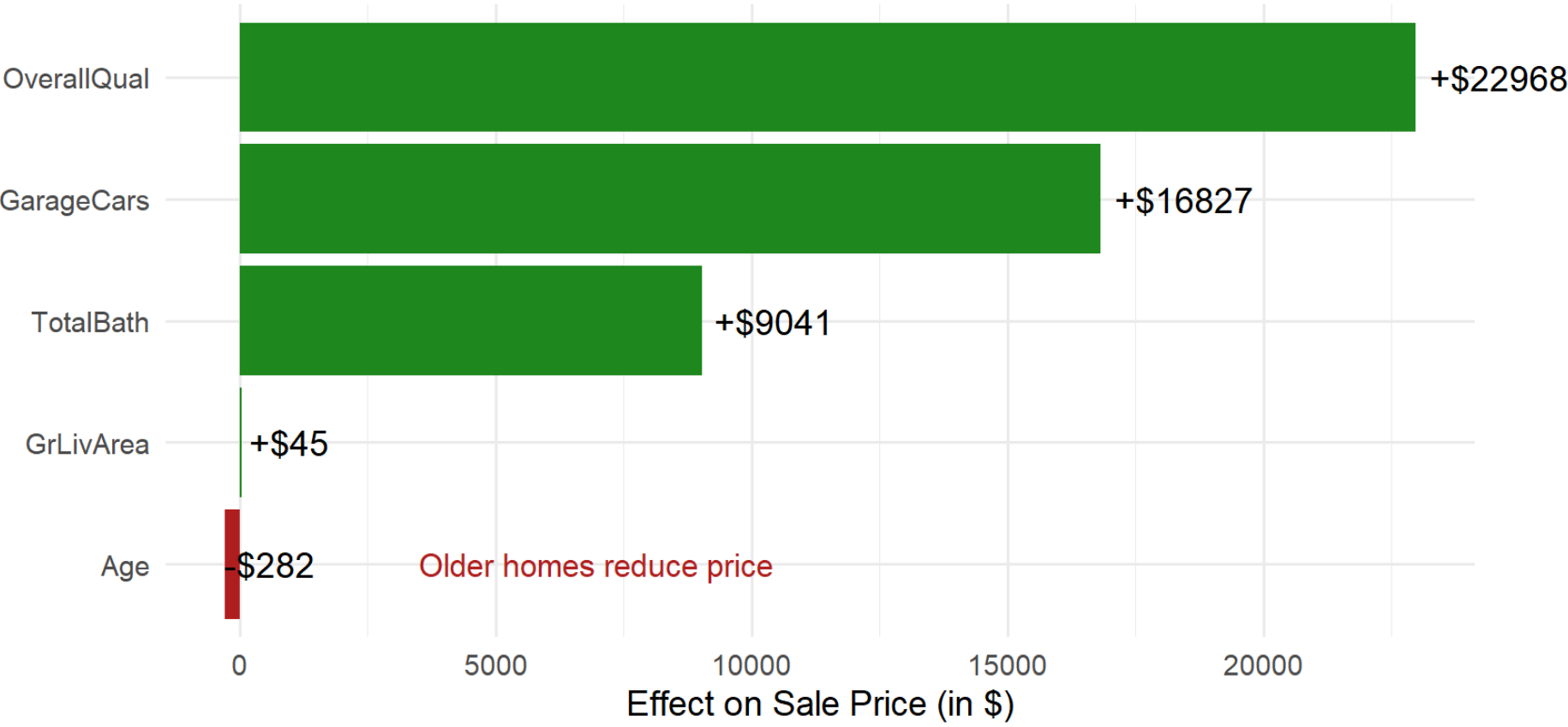
Coefficients:

(Intercept)	GrLivArea	OverallQual	GarageCars	TotalBath	Age
-68048.82	45.42	22967.62	16827.04	9040.93	-281.62

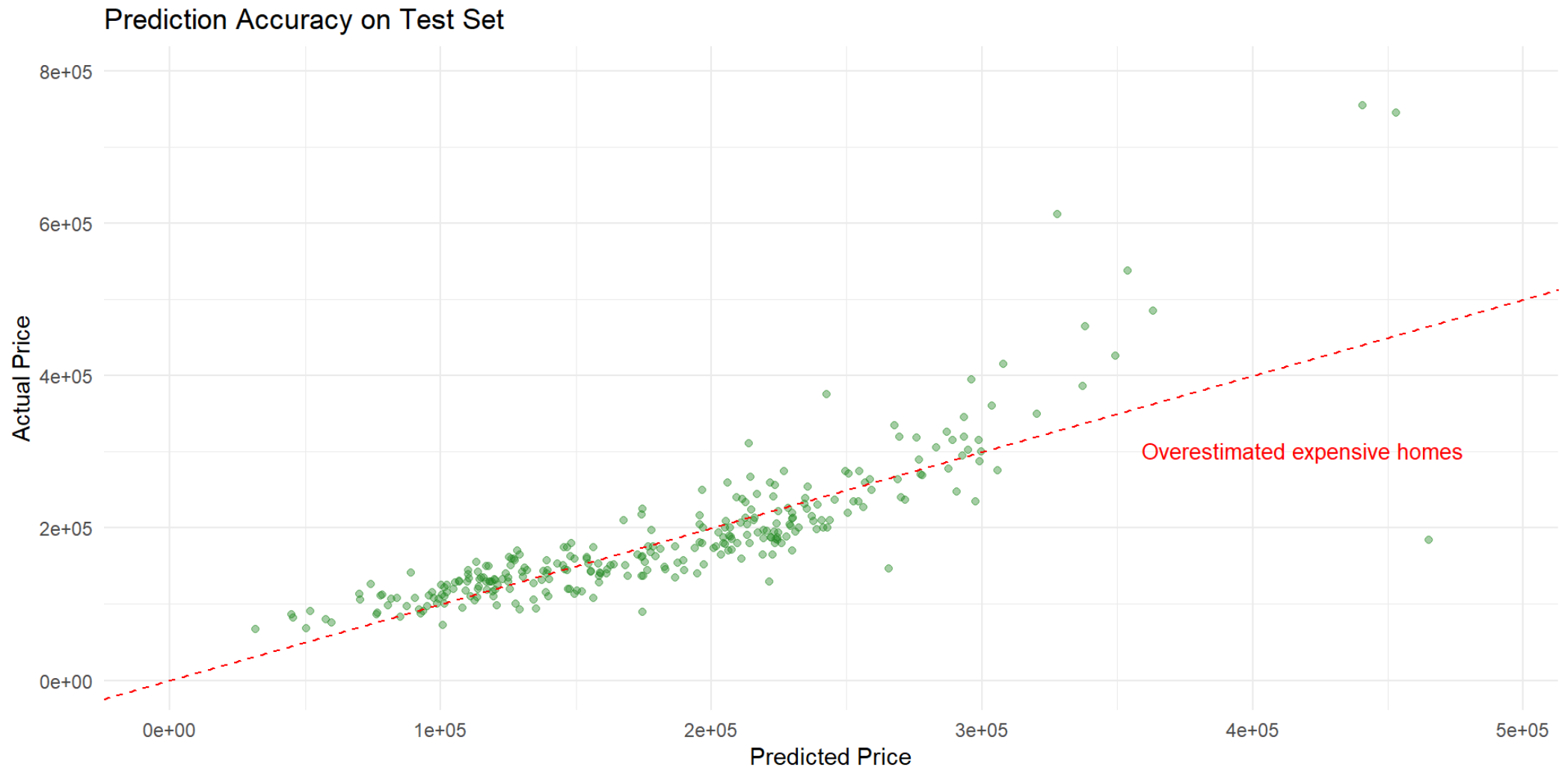
What Drives House Prices?

Quality matters most!

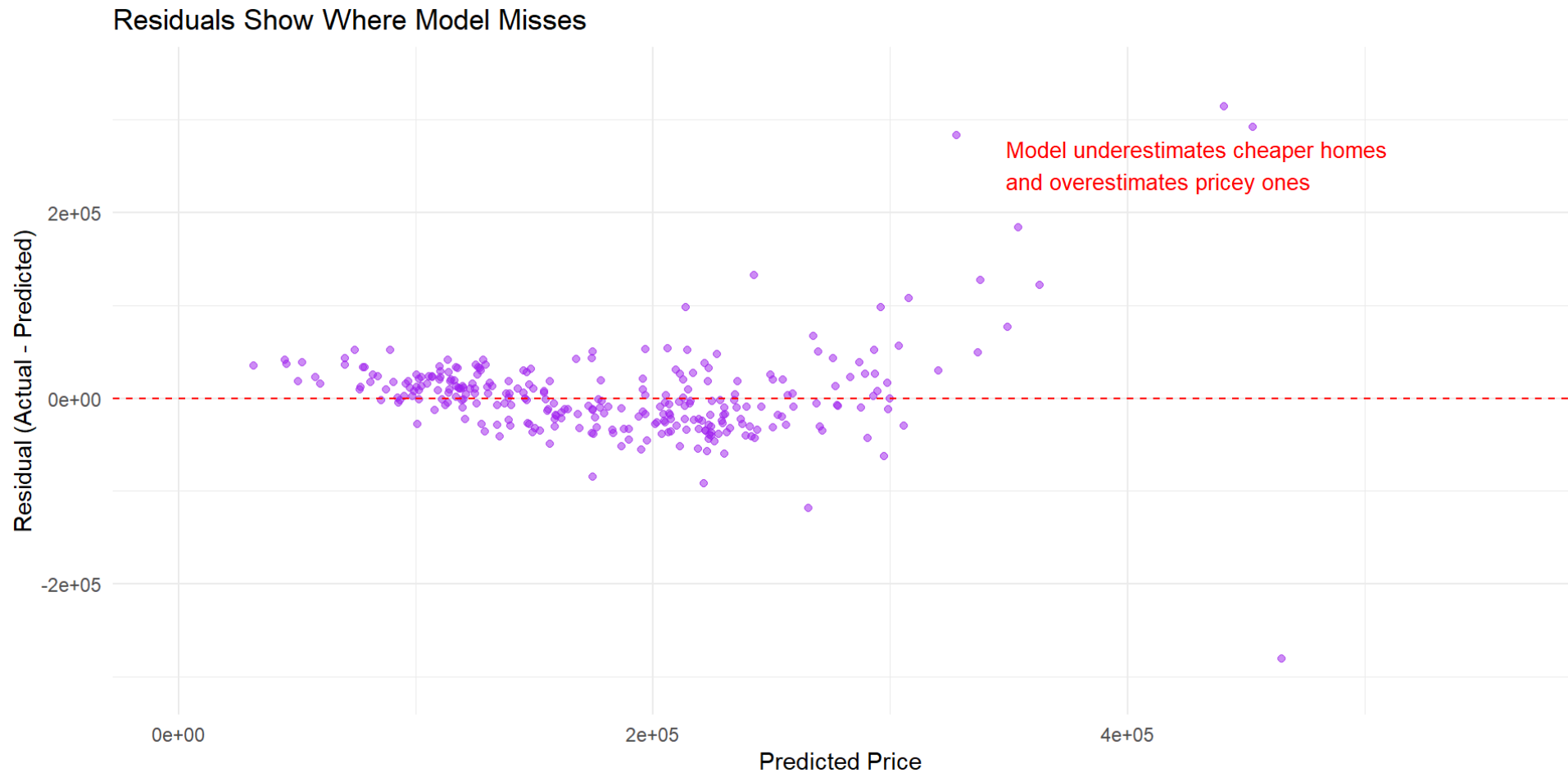
Impact of Each Variable on Sale Price (Based on Linear Model)



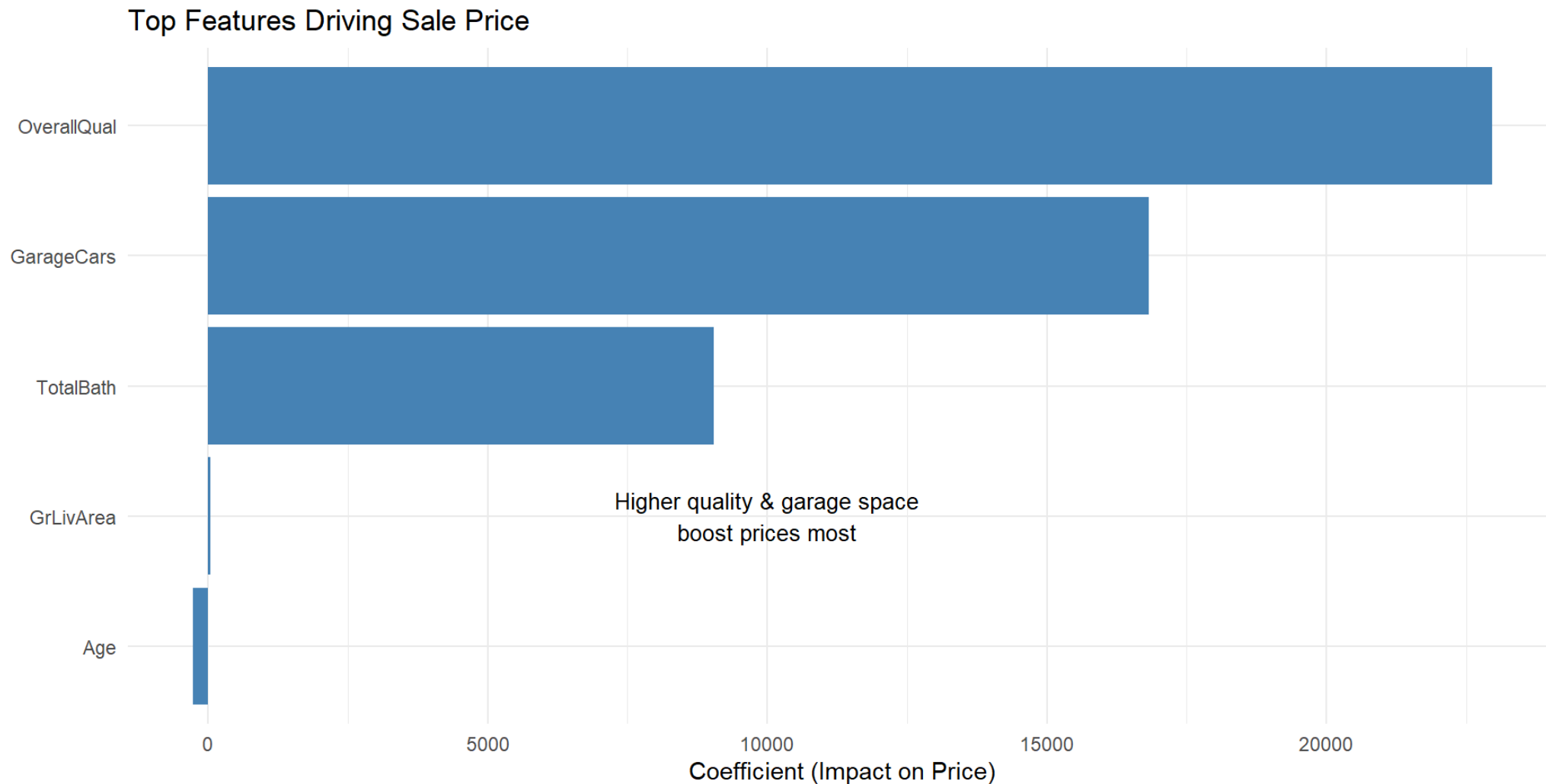
Predictions Closely Follow Actual Prices



Where the Model Struggles: Prediction Errors



Which Factors Matter Most?



Model Explains 71% of the Variability in Housing Prices

Final Model Metrics

RMSE: \$48798

R²: 0.71